Two ghost-path examples for the complete graphic I_4^1

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These examples use the notation and equations of the accompanying paper. They illustrate the complete physical gain estimates when parameter scales are very widely separated; no proof of the main cyclicity theorem depends on them.

1 Two rapidly degenerating ghost paths

These two coefficient paths illustrate why the complete physical gain must be estimated before its individual factors are simplified. In the quadratic chart, large common logarithms cancel in the exact centre matching. In the compact cubic chart, the positive bottleneck forces the negative lip extremely close to its simple root, and the resulting endpoint power remains in the output gain. Both paths use the affine input and divided closing of Lemma 2.2, on the original buffered carrier below a fixed positive raw cap. Their actual centre maps and completed gain quotient are those of Proposition 5.15. We denote the circular constant by π_{circ} when a coefficient π_0 occurs in the same formula.

1.1 A quadratic path with an exponentially small chart ratio

Fix π_0 strictly inside the retained ghost band at $k = 0$, so

$$\pi_\flat < \pi_0 < \sqrt{12}, \quad \pi_0 \le \Pi, \qquad -\Delta_0 < \Delta(0, \pi_0) < 0.$$

Retain a compact raw interval $\mathcal{A} \Subset [d_0, a_0)$, where a_0 is the original positive outward reserve of Corollary J.2.

Proposition 1.1. *For $g \downarrow 0$, take*

$$\begin{aligned} k &= e^{-1/g^2}, \qquad A = 1, \quad C = 1 + g, \quad \mu_3 = -g - g^2, \\ \mu_2 &= (3 + \pi_0 k)g^2, \qquad \mu_1 = -(2 + \pi_0 k + k^2)g^3. \end{aligned} \tag{1.1}$$

For every sufficiently small $g > 0$, the lips with $a \in \mathcal{A}$, $u = a/(gk)$, give actual inputs of the original carrier. Their matched negative radii $R = kr$ satisfy, uniformly on $\mathcal{A}$,

$$R \longrightarrow R_{\pi_0} := \frac{3e^{3D_{\pi_0}}}{1 + e^{3D_{\pi_0}}}, \qquad D_{\pi_0} = \frac{\pi_0 \pi_{\mathrm{circ}}}{3\sqrt{12 - \pi_0^2}}. \tag{1.2}$$

Their raw central heights obey

$$\log H(a) = -\frac{1}{3g^3} - \frac{2}{3g^2} + O(g^{-1} + |\log g|). \tag{1.3}$$

The actual finite channel has $K_{\mathrm{fin}} > 0$ and $m \to 0$. The completed physical derivative satisfies $0 < \Pi' \le C(d_0)g^3$ on these inputs.

Proof. The coefficients tend to θ_0, and the small transverse zero is exactly $\gamma = g$. Substitution in (2.5) gives

$$P_\theta(g+t) = -M(t), \qquad M(t) = t^3 + 3gt^2 - \pi_0 k g^2 t + k^2 g^3.$$

Thus these are fibres of (G.31). The finite discriminant expansion in Section 2 gives

$$K_{\mathrm{fin}} = C - A + O(|\theta - \theta_0|^2) = g + O(g^2) > 0.$$

The actual multiplier (2.12) consequently tends to zero and eventually satisfies the last multiplier restriction.

Put $S = ku = a/g$. Then $u \geq 2M$, $S \geq \delta$, and $a < a_0$ for sufficiently small g. Corollary J.2 places the positive raw image in the prescribed buffer. Its reserve is fixed using the raw section margin, with $a_0 < d_q/L$; it applies when a is fixed while $S \to \infty$. The inward construction in (G.54)–(G.55) gives a positive actual centre height. Its monotonicity and the original comparison give the unique negative match (G.57). Corollary J.3 constructs its outward continuation in the negative buffer, including the original bridge. The two inward graphs have the same height at $v = g$; scalar uniqueness joins them to one through orbit. The finite closing is therefore evaluated on its prescribed image. The inward estimate on the fixed coefficient band gives $H \leq (gk)^2 C g e^{-c/g}$, and the raw speed at the centre is $k^2 g^3 + \ell(g) H$. Both tend to zero. The orbits therefore satisfy the fixed height cap and speed restriction; their positive height places them above the stopped endpoint.

To determine the matched radius, we keep the coefficient-dependent logarithms in the two action primitives until their difference is taken. Set

$$c = \mathfrak{c}, \quad q = c^{-1}, \quad \omega = \sqrt{-\Delta}, \qquad \mathcal{Q}_+(\xi) = (\xi - d)^2 + \omega^2, \quad \mathcal{Q}_-(r) = (r+d)^2 + \omega^2.$$

Then $P(\xi) = (c + k\xi)\mathcal{Q}_+(\xi)$ and $P_-(r) = (c - kr)\mathcal{Q}_-(r)$. Define

$$\beta_q = \frac{1 + 2g + g(2dk + k^2/c^2)}{c + 2dk + k^2/c^2}, \qquad \rho_q = k(g - \beta_q), \qquad \chi_q = -\rho_q/c^2. \tag{1.4}$$

Direct multiplication yields

$$\begin{aligned} \frac{\xi(1 + 2g + gk\xi)}{P(\xi)} &= \frac{\rho_q}{c + k\xi} + \frac{\beta_q \xi + \chi_q}{\mathcal{Q}_+(\xi)}, \\ \frac{r(1 + 2g - gkr)}{P_-(r)} &= -\frac{\rho_q}{c - kr} + \frac{\beta_q r - \chi_q}{\mathcal{Q}_-(r)}. \end{aligned} \tag{1.5}$$

In particular,

$$\beta_q = \frac{1 + 2g}{3} + O(k), \qquad \rho_q/k = g - \beta_q = -\frac{1 - g}{3} + O(k), \qquad \rho_q/k + \beta_q = g.$$

The last identity is exact.

Let $\mathcal{A}_+(u) = \int_0^u \xi B_+/P \, d\xi$ and $\mathcal{A}_-(r) = \int_0^r v B_-/P_- \, dv$, with the same original centre as base point. Integrating (1.5) gives

$$\begin{aligned} \mathcal{A}_+(u) &= (g - \beta_q) \log \frac{c + ku}{c} + \frac{\beta_q}{2} \log \frac{\mathcal{Q}_+(u)}{q} \\ &\quad + \frac{\chi_q + \beta_q d}{\omega} \left(\arctan \frac{u - d}{\omega} + \arctan \frac{d}{\omega} \right), \\ \mathcal{A}_-(r) &= (g - \beta_q) \log \frac{c - kr}{c} + \frac{\beta_q}{2} \log \frac{\mathcal{Q}_-(r)}{q} \\ &\quad - \frac{\chi_q + \beta_q d}{\omega} \left(\arctan \frac{r + d}{\omega} - \arctan \frac{d}{\omega} \right). \end{aligned} \tag{1.6}$$

For $u = S/k$, $r = R/k$, the common logarithmic term is exactly $\beta_q \log(1/k)$. Subtracting the two primitives gives

$$\begin{aligned}\mathcal{A}_+(S/k) - \mathcal{A}_-(R/k) = (g-\beta_q)\log\frac{c+S}{c-R} \\ + \frac{\beta_q}{2}\log\frac{(S-kd)^2+k^2\omega^2}{(R+kd)^2+k^2\omega^2} \\ + \frac{\chi_q+\beta_q d}{\omega}\left(\arctan\frac{S-kd}{k\omega}+\arctan\frac{R+kd}{k\omega}\right).\end{aligned} \tag{1.7}$$

The coefficient of the remaining large $\log S$ in the combined radial terms is g.

The two inward constructions give the actual centre heights

$$\eta_0^+ = g\ell_0 P(u) e^{-\mathcal{A}_+(u)/g} E_+, \qquad \eta_0^- = g\ell_0 P_-(r) e^{-\mathcal{A}_-(r)/g} E_-, \qquad E_\pm, E_\pm^{-1} \le C.$$

The bounded factors contain the nonlinear ratio integrals and the compact continuations to the centre. Their bounds follow from the action-clock decay in the two inward constructions. Matching these actual heights gives

$$\mathcal{A}_+(u) - \mathcal{A}_-(r) = g\log\frac{P(u)}{P_-(r)} + g\log\frac{E_+}{E_-}. \tag{1.8}$$

The initial polynomial factors in these heights have the exact quotient:

$$\frac{P(S/k)}{P_-(R/k)} = \frac{k^2 - \pi_0 kS + 3S^2 + S^3}{k^2 + \pi_0 kR + 3R^2 - R^3}. \tag{1.9}$$

On a fixed compact subinterval of $0 < R < 3$, its logarithm is $3\log S - \log[R^2(3-R)] + O(S^{-1}+k)$. Thus its contribution to (1.8) is $O(g(1+|\log g|))$. The common terms containing $\log k$ cancel in the exact action and polynomial identities before the estimate is taken, although $g\log(1/k) = g^{-1} \to \infty$ on this path.

For a fixed $R \in (0,3)$, (1.7) tends to

$$D_{\pi_0} + \frac{1}{3}\log\frac{3-R}{R}.$$

Indeed $d \to \pi_0/6$, $\omega \to \sqrt{12-\pi_0^2}/6$, and both arctangents tend to $\pi_{\mathrm{circ}}/2$. This limiting expression has the single zero R_{π_0} in (1.2). Choose fixed $R_{\mathrm{lo}} < R_{\pi_0} < R_{\mathrm{hi}}$ in $(0,3)$. The exact matching residual has opposite signs at these two radii for small g, uniformly on $\mathcal{A}$. Strict monotonicity of the actual negative centre map places its match between them. The expansion is uniform on that compact interval; shrinking the brackets proves (1.2). The limiting gap from the negative root is positive for this fixed π_0.

The positive primitive also gives

$$\mathcal{A}_+(u) = \frac{1+2g}{3}\log(1/k) + g\log S + O(1).$$

Since $H = \ell_0 M(a) e^{-\mathcal{A}_+(u)/g} E_+$ and $M(a) \to a^3$, this proves (1.3). In the scaled centre coordinate,

$$\log\eta_0^+ = -\frac{1}{3g^3} + \frac{4}{3g^2} + O(g^{-1} + |\log g|).$$

Finally retain the two complete exterior gains. With $\nu = 6g/(1+2g)$, $\lambda_g = 2\nu$, $p = \widehat{P}(R)/R^2$, and $b_R = -R^2/2$, the negative gain is

$$\Gamma'_{-,b_R} = c_- g^2 \ell_0^{-1-\nu} g^{-\nu} (gR)^{-\lambda_g} p^{-\nu} \mathcal{K}_-, \qquad 0 < \mathcal{K}_- \le C. \tag{1.10}$$

Its original label change is $\Gamma'_{-,b_r} = k^2\Gamma'_{-,b_R}$, where $b_r = -r^2/2$. The positive input gain in $b_u = -u^2/2$ retains $(gk)^2$. Using the actual centre variations gives

$$\begin{aligned} 0 < \frac{d\Gamma_-}{d\log\eta_0^-} &\le Cc_-g^3 g^{-\nu-\lambda_g} R^{2-\lambda_g} p^{1-\nu} \le Cc_-g^3, \\ \frac{d\Gamma_+}{d\log\eta_0^+} &\ge c(d_0)c_+a^3. \end{aligned} \tag{1.11}$$

The bounded negative factor follows from the upper density estimate (G.59). Its root dwell has nonpositive density, as shown in (G.58); it does not require an inverse bound on $\mathcal{K}_-$. The exact quotient of (1.11) is the completed physical Π'. This proves its stated bound. If the exterior gain is factored differently, the term $k^{-\lambda_g} = \exp(12/[g(1+2g)])$ is unbounded on this path; the exact label change and centre variation account for it before the quotient is estimated. □

1.2 A compact cubic path with a very small positive bottleneck

We write $\tau_{\mathrm{c}} = \tau_{\mathrm{cub}}$ for the raw cubic scale and κ_{c} for its positive ghost gap. Choose a compact block with $U_0 \ge 1$ and $G_0 > 1/3$. For small $\kappa_{\mathrm{c}} > 0$, set

$$\begin{aligned} \kappa_{\mathrm{c}} = e^{-1/\tau_{\mathrm{c}}^2}, \qquad c_{\mathrm{c}} = (1+\kappa_{\mathrm{c}}^2)^{-1}, \qquad \alpha_{\mathrm{c}} = 1, \\ U_{\mathrm{c}} = 2c_{\mathrm{c}} - c_{\mathrm{c}}^{-1}, \qquad G_{\mathrm{c}} = (c_{\mathrm{c}} - 2)/3. \end{aligned} \tag{1.12}$$

These coefficients belong to the block for sufficiently small τ_{c}, and $U_{\mathrm{c}} \ge 0$. Their axis polynomial is

$$P_{\mathrm{c}}(v) = (v + c_{\mathrm{c}})\{(v-1)^2 + \kappa_{\mathrm{c}}^2\}, \qquad \Delta_{\mathrm{c}} = -\kappa_{\mathrm{c}}^2 < 0. \tag{1.13}$$

There is a fixed raw reserve $a_{\mathrm{res}} > 0$ inside the positive window for this path. To see this, at $\tau_{\mathrm{c}} = 0$ consider the positive raw graph with $M(t) = t^3$, initial height $z(a) = \ell_0 a^3$, and the organizing passive coefficients. The near-lip comparison of Lemma G.3 puts z/t^2 in a fixed positive strip at $t = 2a$. The subsequent equation is $t(z/t^2)' = 1 - 2z/t^2 + O(t)$, with a negative normal diagonal on that strip. Its fixed-port value converges to the organizing strong graph as $a \downarrow 0$. The strict window margin therefore admits all these limiting raw images for $0 < a \le a_{\mathrm{res}}$, after decreasing a_{res}. Ordinary dependence on the regular raw interval from a fixed $a > 0$ to the port preserves this margin for small τ_{c}, uniformly on compact raw subintervals. Choose the auxiliary d_0, after the original windows and reserves, so that $2d_0 < a_{\mathrm{res}}$, while retaining the small-error requirements of its construction. We may therefore use $a \in [d_0, 2d_0]$ within the originally prescribed physical carrier.

Proposition 1.2. *The path* (1.12) *is realized in the full coefficient germ by*

$$\begin{aligned} A = 1, \quad C = 1 + \tau_{\mathrm{c}}, \quad \gamma = \tau_{\mathrm{c}} G_{\mathrm{c}}, \quad \mu_3 = -\gamma - \gamma^2, \\ \mu_2 = \tau_{\mathrm{c}}^2(U_{\mathrm{c}} + 3G_{\mathrm{c}}^2), \qquad \mu_1 = -\tau_{\mathrm{c}}^3(1 + U_{\mathrm{c}}G_{\mathrm{c}} + 2G_{\mathrm{c}}^3). \end{aligned} \tag{1.14}$$

For sufficiently small $\tau_{\mathrm{c}} > 0$, the lips with $a \in [d_0, 2d_0]$, $u = a/\tau_{\mathrm{c}}$, are original admitted inputs. If $X = c_{\mathrm{c}} - r$ is their matched negative root distance, then uniformly on this interval

$$\kappa_{\mathrm{c}} \log(1/X) \longrightarrow 2\pi_{\mathrm{circ}}. \tag{1.15}$$

Their raw central height satisfies

$$\log H(a) = -\frac{\pi_{\mathrm{circ}} B(1)}{(1+c_{\mathrm{c}})\tau_{\mathrm{c}}\kappa_{\mathrm{c}}} + O(\tau_{\mathrm{c}}^{-1} + |\log\tau_{\mathrm{c}}|), \tag{1.16}$$

so $\tau_{\mathrm{c}}\kappa_{\mathrm{c}} \log H(a) \to -\pi_{\mathrm{circ}}/2$. The finite channel has $K_{\mathrm{fin}} > 0$, $m \to 0$, and the full physical derivative obeys $0 < \Pi' \le C(d_0)\tau_{\mathrm{c}}^3$.

Proof. Translation of the raw polynomial gives

$$P_\theta(\gamma + t) = -M_{\mathrm{c}}(t), \qquad M_{\mathrm{c}}(t) = t^3 + 3\tau_{\mathrm{c}} G_{\mathrm{c}} t^2 - \tau_{\mathrm{c}}^2 U_{\mathrm{c}} t + \tau_{\mathrm{c}}^3.$$

The coefficients thus give exactly (4.10)–(4.11). Moreover $K_{\mathrm{fin}} = \tau_{\mathrm{c}} + O(\tau_{\mathrm{c}}^2) > 0$, so the actual finite multiplier tends to zero. The positive port images are buffered by the fixed reserve above. The inward construction (4.23) and (4.24) gives separate actual positive centre maps, with bounded nonlinear memory factors. Write their actions as $\mathcal{A}_{\mathrm{c},+}$ and $\mathcal{A}_{\mathrm{c},-}$, based at the original centre.

For $u = a/\tau_{\mathrm{c}}$, direct integration across the bottleneck gives

$$\mathcal{A}_{\mathrm{c},+}(u) = \frac{\pi_{\mathrm{circ}} B(1)}{(1 + c_{\mathrm{c}})\kappa_{\mathrm{c}}} + O(1 + \tau_{\mathrm{c}} \log u). \tag{1.17}$$

For completeness, put $h(v) = vB(v)/(v + c_{\mathrm{c}})$. On a fixed symmetric interval about $v = 1$, subtract $h(1) + h'(1)(v - 1)$. The linear term integrates to zero against $((v - 1)^2 + \kappa_{\mathrm{c}}^2)^{-1}$, and the quadratic remainder has bounded integral. Outside this interval the denominator has a fixed margin; for large v, $h(v) = O(1 + \tau_{\mathrm{c}} v)$, giving the remaining $O(1 + \tau_{\mathrm{c}} \log u)$ bound. Thus $\kappa_{\mathrm{c}} \mathcal{A}_{\mathrm{c},+} \to \pi_{\mathrm{circ}}/2$.

At a fixed negative lip $r_0 < c_{\mathrm{c}}$, the negative action is bounded. The positive centre height from (1.17) is smaller than its negative centre height for small τ_{c}. The actual image (4.25) therefore gives a unique match, before an observer is attached. Its radius tends to c_{c}: on every fixed complement of that root, the negative action and polynomial logarithm stay bounded, whereas the positive action diverges. The negative root-collar construction in Lemma G.4 gives a positive-time outward passage on a fixed buffered invariant chord and continues it to the original negative raw section. It applies for arbitrarily small X. This proves negative output and closing admission. Equality at the same centre joins the two graphs by scalar uniqueness.

Both initial polynomial factors occur in the exact centre matching, so

$$\mathcal{A}_{\mathrm{c},+}(u) - \mathcal{A}_{\mathrm{c},-}(r) = \tau_{\mathrm{c}} \log \frac{P_{\mathrm{c}}(u)}{P_{\mathrm{c}}(-r)} + O(\tau_{\mathrm{c}}). \tag{1.18}$$

Put

$$L_-(r) = (r + 1)^2 + \kappa_{\mathrm{c}}^2, \qquad a_- := \frac{c_{\mathrm{c}} B(-c_{\mathrm{c}})}{L_-(c_{\mathrm{c}})} \longrightarrow \frac{1}{4}.$$

The simple-root action and the source polynomial give, uniformly in a fixed negative collar,

$$\mathcal{A}_{\mathrm{c},-}(r) = a_- \log(1/X) + O(1), \qquad \log \frac{P_{\mathrm{c}}(u)}{P_{\mathrm{c}}(-r)} = 3 \log u + \log(1/X) + O(1).$$

Inserting these in (1.18) yields

$$(a_- + \tau_{\mathrm{c}}) \log(1/X) = \mathcal{A}_{\mathrm{c},+}(u) + O(1 + \tau_{\mathrm{c}} \log u).$$

This proves (1.15). Also $H = \ell_0 M_{\mathrm{c}}(a) e^{-\mathcal{A}_{\mathrm{c},+}(u)/\tau_{\mathrm{c}}} E_+$, with $E_+, E_+^{-1} \le C$ and $M_{\mathrm{c}}(a) \to a^3$. Equation (1.17) proves (1.16). The height is positive and below the fixed cap. The raw speed at the centre is $\tau_{\mathrm{c}}^3 + \ell(\gamma) H$, which tends to zero, so the original speed restriction holds as well.

The root distance is too small to discard its power in the negative output gain. We keep that power while forming the physical derivative. With

$$\lambda_G = \frac{12 \tau_{\mathrm{c}} G_{\mathrm{c}}}{1 + 2\tau_{\mathrm{c}} G_{\mathrm{c}}}, \qquad \beta_- = \frac{\tau_{\mathrm{c}} L_-(c_{\mathrm{c}})}{c_{\mathrm{c}} B(-c_{\mathrm{c}})} = \frac{\tau_{\mathrm{c}}}{a_-},$$

the complete collar gain retains

$$\Gamma'_{-,b_-} = \tau_{\mathrm{c}}^{2-\lambda_G} \mathcal{U}_{\mathrm{c}} \, \tau_{\mathrm{c}}^{\beta_-} X^{\beta_-} K_{\mathrm{c}}(X, X^{\beta_-}), \qquad b_- = -r^2/2. \tag{1.19}$$

Here $\mathcal{U}_{\mathrm{c}}$ and K_{c} are bounded positive units; the fixed output bridge and divided closing are included in their bounded terminal factors. Equation (1.15) gives

$$\beta_{-}\log(1/X)\sim\frac{8\pi_{\mathrm{circ}}\tau_{\mathrm{c}}}{\kappa_{\mathrm{c}}}\longrightarrow\infty,\qquad X^{\beta_{-}}\longrightarrow 0.$$

The input gain in $b_{+}=-u^{2}/2$ satisfies $\Gamma'_{+,b_{+}}\asymp_{d_0}\tau_{\mathrm{c}}^{2}$ by (4.22). The physical quotient is exactly

$$\Pi'=\frac{\Gamma'_{-,b_{-}}}{\Gamma'_{+,b_{+}}}\frac{B_{+}P_{-}}{B_{-}P_{+}}\frac{\Lambda_{+}}{\Lambda_{-}}.\tag{1.20}$$

Since $\lambda_G=O(\tau_{\mathrm{c}})$, $\tau_{\mathrm{c}}^{-\lambda_G}$ is bounded. The factors $\tau_{\mathrm{c}}^{\beta_{-}}$ and $X^{\beta_{-}}$ are at most one. The input polynomial satisfies $P_{+}(u)\geq c(d_0)\tau_{\mathrm{c}}^{-3}$, whereas $P_{-}\leq C$ and the two centre variation factors are units. This proves $0<\Pi'\leq C(d_0)\tau_{\mathrm{c}}^{3}$, with the original divided closing and all exterior gains retained. □